CSE 270 Sports Analytics

Homework 3

Name: _____

Competitive Balance

Problem 1 (50 points)

Noll-Scully number

1. Get the final tables for all countries in the dataset *f_data_sm* for all seasons(for each country). (10 points)
2. Calculate the wining percentage(ratio) for all the teams in all seasons. (Consider draws as half wins). (2 points)
3. Calculate the Noll-Scully number for each season and country using the winning ratio. (3 points)
4. Visualize the results grouped by seasons and countries. Make sure to include meaningful title and axis names.
   (You can use facets) (10 points)

5.Interpret the visualization (5 points)

C5 competitive balance

6.Calculate the C5 index for the number of goals scored by the top 5 teams in each country for each season. You should use the dataset from Problem 1.1. (5 points)

7. Visualize the results for all countries and seasons. (Can be the same structure as in 1.4) (10 points)
8. Interpret the results (5 points)

Individual Performance (50 points)

Problem 2

1. Load the dataset *nba_players* from the library *SportsAnalytics270* and filter it to only include seasons starting from 1992.
   Remove the columns blanl and blank2 and drop missing values or nas. (2 points)
2. Clean the variable Pos from players with double positions, for example C-PF should be turned into C. (5 points)
3. Convert the vaiable Pos into a factor and remove the columns X, Year, Player and Tm. (3 points)

4. Divide the dataset into train and test datasets. (You can use dplyr or any other sampling method) (10 points)

5. Build a classifier model to predict the position of the player based on the playing attributes. Use the training set for fitting the model. (20 points) You can use multinomial regression for that

6. Predict the position of the players for the test dataset. (5 points)

7. Interpret the predictive power of the model. (5 points)